

Assignment 2

Docker Compose for Multi-Container App

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Task 1.1

The screenshot shows the Visual Studio Code interface with a project named 'my-node-app'. The Explorer sidebar on the left shows the file structure: 'node_modules', 'Dockerfile', 'package-lock.json', 'package.json', and 'server.js'. The 'server.js' file is selected and its content is displayed in the main editor. The code is a simple Express.js server that listens on port 3000 and responds with 'Hello, Docker!' to any GET request. The bottom panel shows the 'TERMINAL' tab with the following commands and output:

```
PS F:\SEM_7\Devops_Lab\my-node-app> npm init -y
PS F:\SEM_7\Devops_Lab\my-node-app> npm install express
* History restored

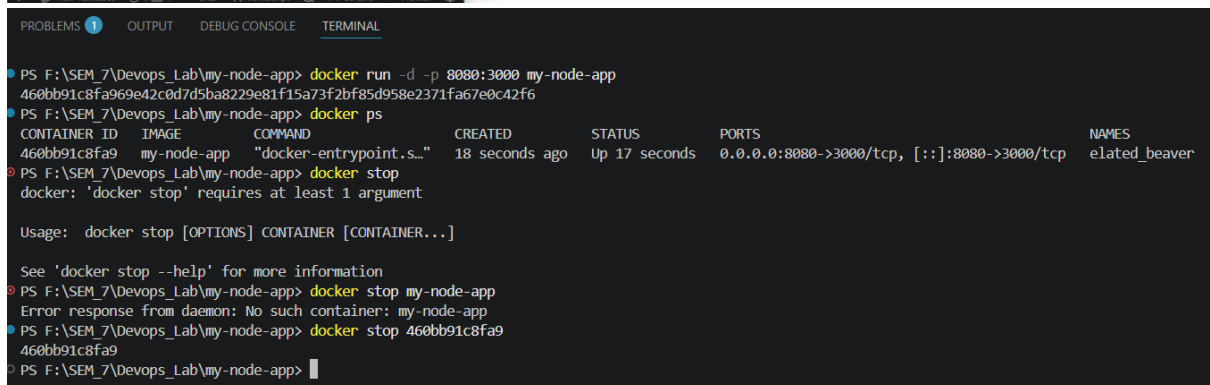
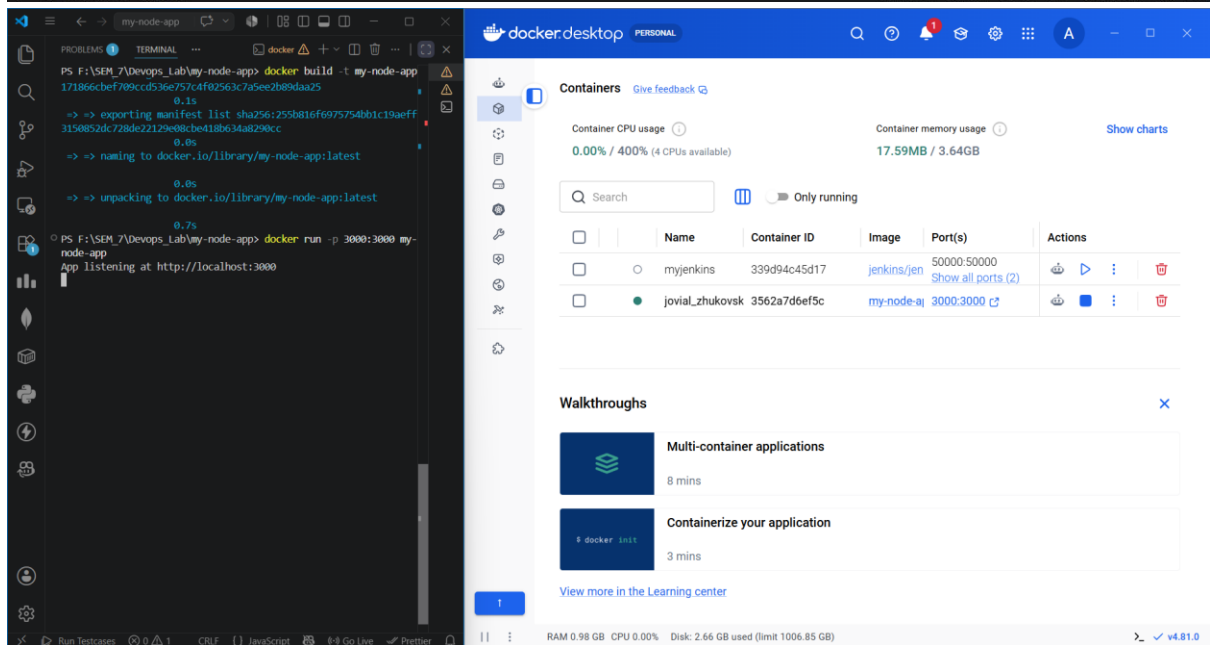
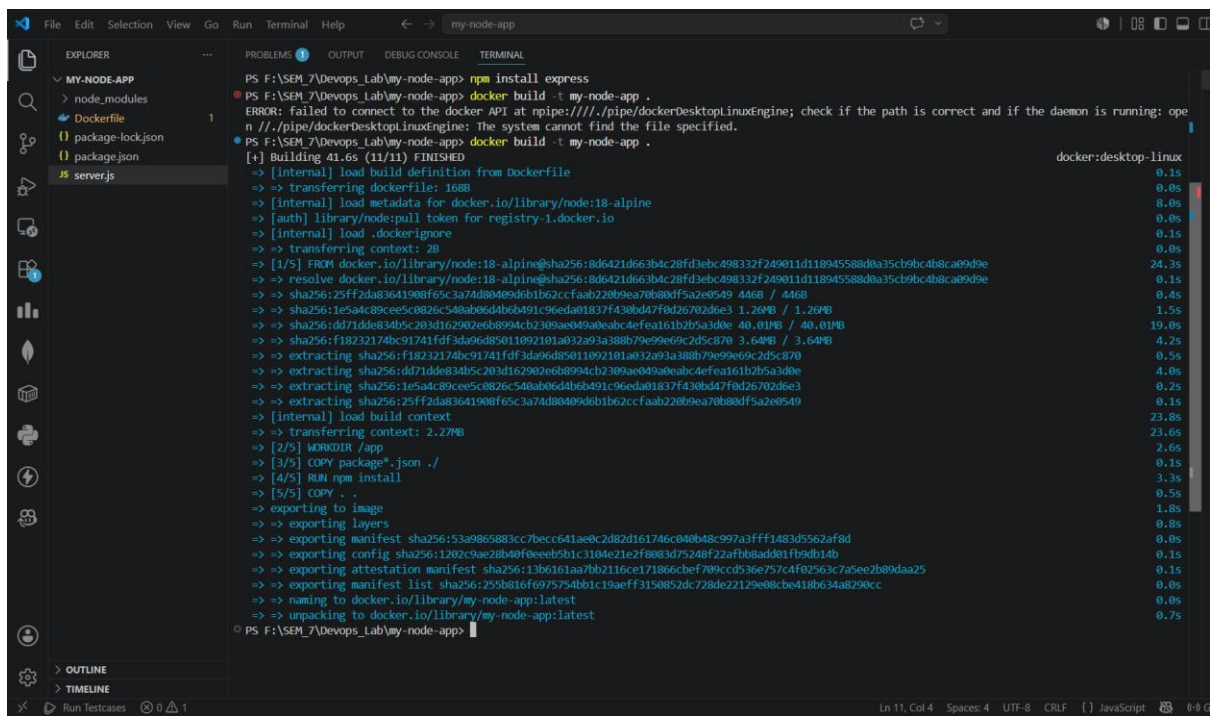
PS F:\SEM_7\Devops_Lab\my-node-app>
* History restored

● PS F:\SEM_7\Devops_Lab\my-node-app> npm install express

added 67 packages, and audited 68 packages in 52s

25 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```



Gordon

Containers

Images

Logs

Volumes

Kubernetes

Builds

Models

Docker Hub

MCP Toolkit

Extensions

Containers

Give feedback

Container CPU usage

No containers are running.

Container memory usage

No containers are running.

Show charts

Search

Only show running containers

		Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	☐	myjenkins	339d94c45d17	jenkins/jenkins:its-jdk17	50000:50000 Show all ports (2)	N/A	6 days ago	
☐	☐	jovial_zhukovsky	3562a7d6ef5c	my-node-app	3000:3000	N/A	10 minutes ago	
☐	☐	elated_beaver	460bb91c8fa9	my-node-app	8080:3000	N/A	3 minutes ago	

Showing 3 items

Walkthroughs

Multi-container applications

8 mins

Containerize your application

3 mins

View more in the Learning center

Engine running

RAM 0.93 GB CPU 0.00% Disk: 2.66 GB used (limit 1006.85 GB)

Terminal

v4.81.0

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PS F:\SEM_7\Devops_Lab\my-node-app> docker stop 460bb91c8fa9

PS F:\SEM_7\Devops_Lab\my-node-app> docker ps -a

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
460bb91c8fa9	my-node-app	"docker-entrypoint.s..."	2 minutes ago	Exited (137) About a minute ago		elated_beaver
3562a7d6ef5c	my-node-app	"docker-entrypoint.s..."	10 minutes ago	Exited (137) 3 minutes ago		jovial_zhukovsky
339d94c45d17	jenkins/its-jdk17	"/usr/bin/tini -- /u..."	7 days ago	Exited (255) 17 minutes ago	0.0.0.0:8080->8080/tcp, 0.0.0.0:50000->50000/tcp	myjenkins

PS F:\SEM_7\Devops_Lab\my-node-app> docker inspect 460bb91c8fa9

[

{

{

"Id": "460bb91c8fa9e9e42c0d7d5ba8229e81f15a73f2bf85d958e2371fa67e0c42f6",

"Created": "2026-07-22T07:55:12.883043052Z",

"Path": "docker-entrypoint.sh",

"Args": [

"node",

"server.js"

],

"State": {

"Status": "exited",

"Running": false,

"Paused": false,

"Restarting": false,

"OOMKilled": false,

"Dead": false,

"Pid": 0,

"ExitCode": 137,

"Error": "",

"StartedAt": "2026-07-22T07:55:13.053948847Z",

"FinishedAt": "2026-07-22T07:56:34.911628336Z"

},

"Image": "sha256:255b816f6975754bb1c19aeff3150852dc728de22129e08cbe418b634a8290cc",

"ResolvConfPath": "/var/lib/docker/containers/460bb91c8fa9e9e42c0d7d5ba8229e81f15a73f2bf85d958e2371fa67e0c42f6/resolv.conf",

"HostnamePath": "/var/lib/docker/containers/460bb91c8fa9e9e42c0d7d5ba8229e81f15a73f2bf85d958e2371fa67e0c42f6/hostname",

"HostsPath": "/var/lib/docker/containers/460bb91c8fa9e9e42c0d7d5ba8229e81f15a73f2bf85d958e2371fa67e0c42f6/hosts",

"LogPath": "/var/lib/docker/containers/460bb91c8fa9e9e42c0d7d5ba8229e81f15a73f2bf85d958e2371fa67e0c42f6/460bb91c8fa9e9e42c0d7d5ba8229e81f15a73f2bf85d958e2371fa67e0c42f6-json.log",

"Name": "/elated_beaver",

"RestartCount": 0,

"Driver": "overlayfs",

"Platform": "linux",

"MountLabel": "",

"ProcessLabel": "",

"AppArmorProfile": ""

}

}

}

}

Run Testcases

Ln 11, Col 4 - Spaces: 4 UTF-8 CRLF JavaScript 8-8 Go Live Print

my-node-app

Welcome

server.js

Dockerfile 1

```
1 const express = require('express');
2 const app = express();
3 const port = 3000;
4
5 app.get('/', (req, res) => {
6   res.send('Hello, Docker!');
7 });
8
9 app.listen(port, '0.0.0.0', () => {
10   console.log(`App listening at http://localhost:${port}`);
11 });
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

powershell

```
PS F:\SEM_7\Devops_Lab\my-node-app> docker rm
Usage: docker rm [OPTIONS] CONTAINER [CONTAINER...]

See 'docker rm --help' for more information
PS F:\SEM_7\Devops_Lab\my-node-app> docker rm .
Error response from daemon: page not found
PS F:\SEM_7\Devops_Lab\my-node-app> docker rm 46ebb91c8fa9
46ebb91c8fa9
PS F:\SEM_7\Devops_Lab\my-node-app> docker rm 3562a7d6ef5c
3562a7d6ef5c
PS F:\SEM_7\Devops_Lab\my-node-app> 
```

Run Testcases 0 1 Lin 11, Col 4 Spaces 4 UTF-8 CRLF JavaScript

docker desktop PERSONAL

Containers Give feedback

Container CPU usage Container memory usage Show charts

No containers are running. No containers are running.

Search Only running

	Name	Container ID	Image	Port(s)	Actions
	myjenkins	339d94c45d17	jenkins/jen	50000:50000 Show all ports (2)	

Walkthroughs

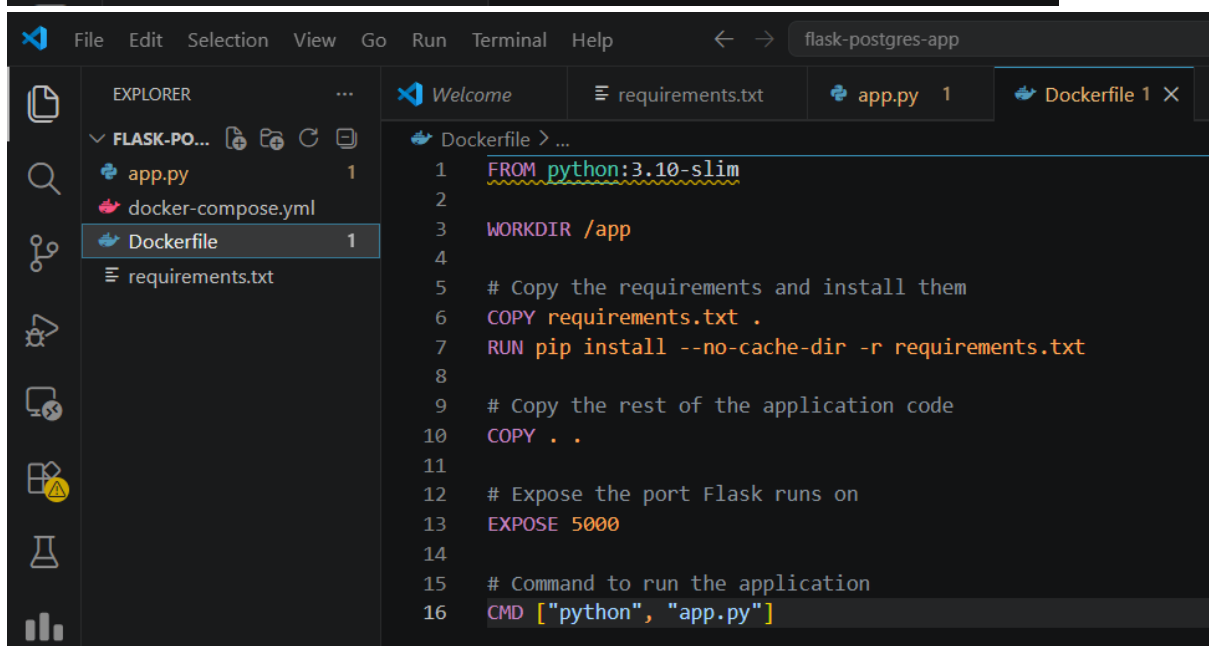
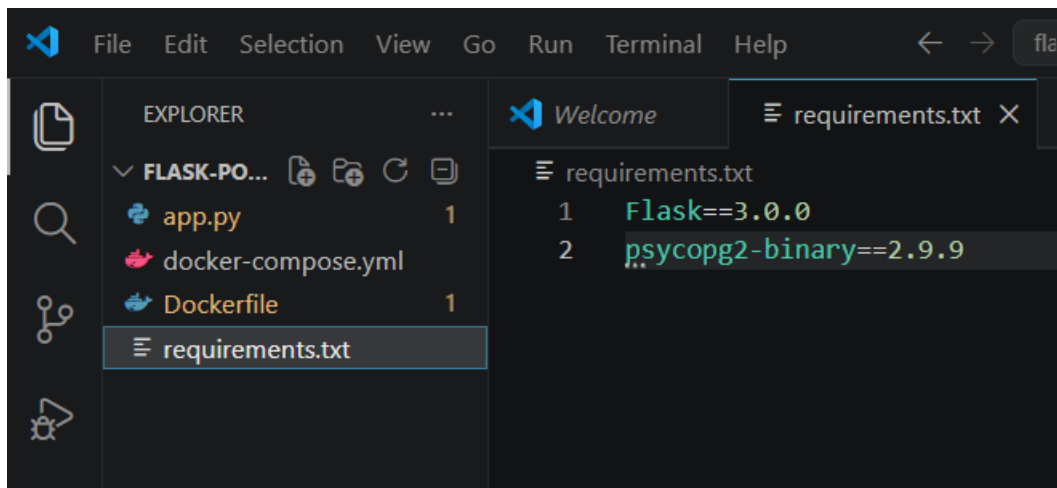
Multi-container applications
8 mins

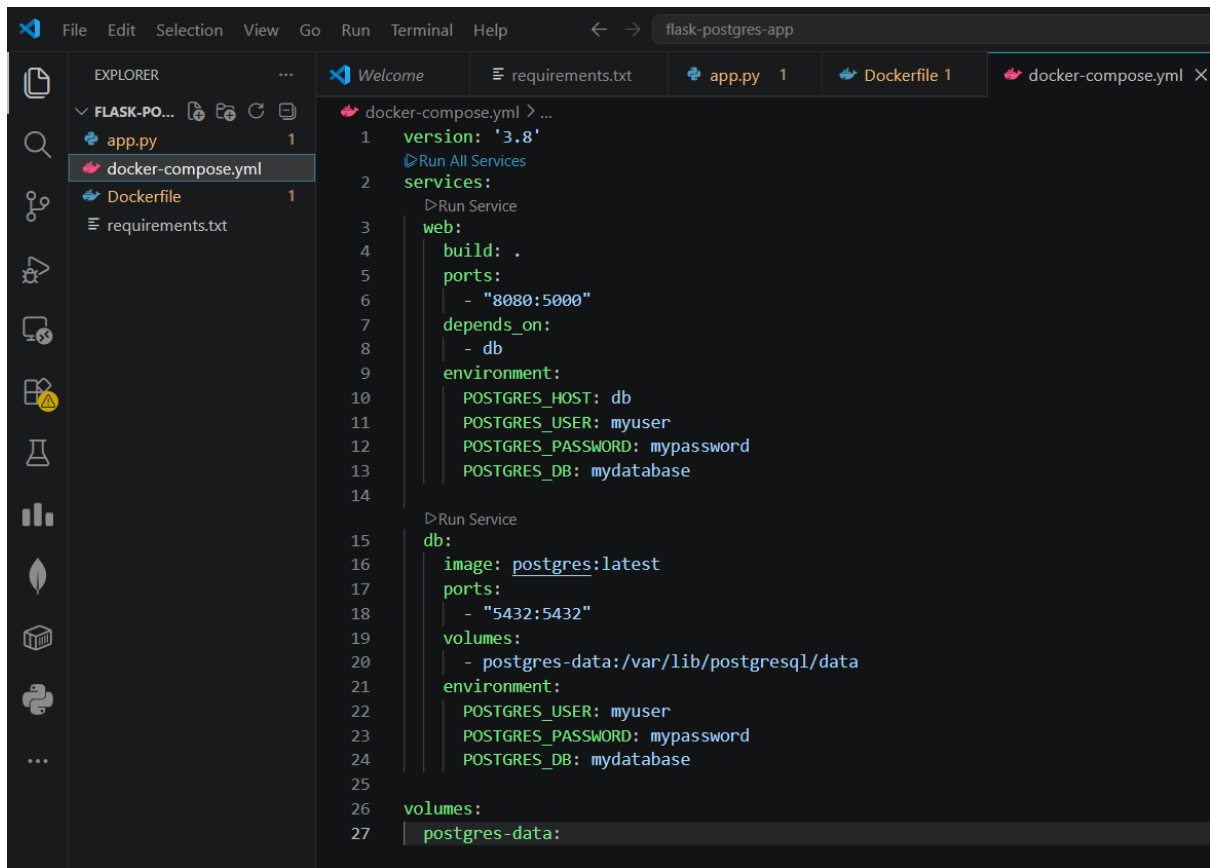
Containerize your application
3 mins

View more in the Learning center

RAM 1.00 GB CPU 0.50% Disk: 2.66 GB used (limit 1006.85 GB) v4.81.0

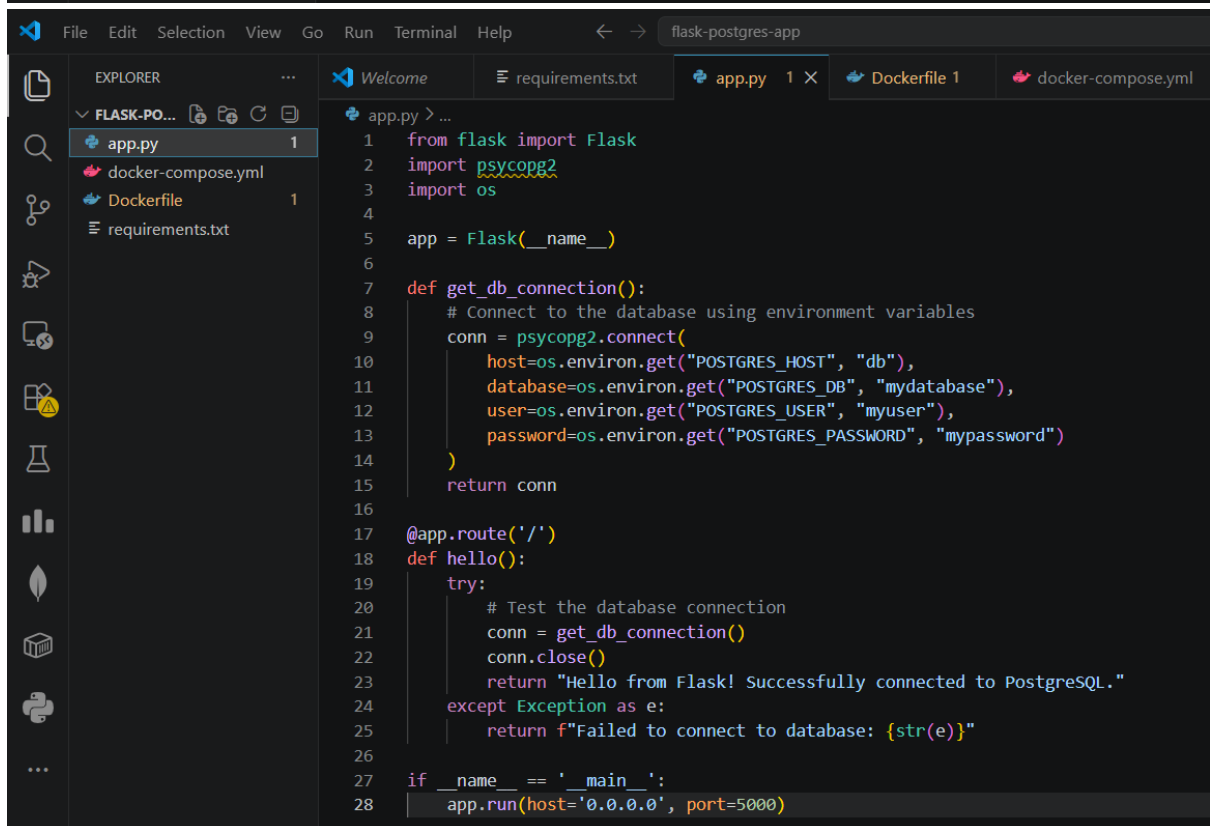
Task 1.2





This screenshot shows the Visual Studio Code editor with the `docker-compose.yml` file open. The Explorer sidebar on the left shows the project structure with `app.py`, `docker-compose.yml`, `Dockerfile`, and `requirements.txt`. The main editor area displays the following YAML configuration:

```
1 version: '3.8'
2 services:
3   web:
4     build: .
5     ports:
6       - "8080:5000"
7     depends_on:
8       - db
9     environment:
10      POSTGRES_HOST: db
11      POSTGRES_USER: myuser
12      POSTGRES_PASSWORD: mypassword
13      POSTGRES_DB: mydatabase
14
15   db:
16     image: postgres:latest
17     ports:
18       - "5432:5432"
19     volumes:
20       - postgres-data:/var/lib/postgresql/data
21     environment:
22      POSTGRES_USER: myuser
23      POSTGRES_PASSWORD: mypassword
24      POSTGRES_DB: mydatabase
25
26 volumes:
27   postgres-data:
```



This screenshot shows the Visual Studio Code editor with the `app.py` file open. The Explorer sidebar on the left shows the project structure. The main editor area displays the following Python code:

```
1 from flask import Flask
2 import psycopg2
3 import os
4
5 app = Flask(__name__)
6
7 def get_db_connection():
8     # Connect to the database using environment variables
9     conn = psycopg2.connect(
10         host=os.environ.get("POSTGRES_HOST", "db"),
11         database=os.environ.get("POSTGRES_DB", "mydatabase"),
12         user=os.environ.get("POSTGRES_USER", "myuser"),
13         password=os.environ.get("POSTGRES_PASSWORD", "mypassword")
14     )
15     return conn
16
17 @app.route('/')
18 def hello():
19     try:
20         # Test the database connection
21         conn = get_db_connection()
22         conn.close()
23         return "Hello from Flask! Successfully connected to PostgreSQL."
24     except Exception as e:
25         return f"Failed to connect to database: {str(e)}"
26
27 if __name__ == '__main__':
28     app.run(host='0.0.0.0', port=5000)
```

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL powershell + - - - - -
PS F:\SEM_7\Devops_Lab\flask-postgres-app> docker-compose up -d --build
time="2026-07-22T13:56:43+05:30" level=warning msg="F:\SEM_7\Devops_Lab\flask-postgres-app\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 16/16
  ✓ Image postgres:latest pulled 25.2s
#1 [internal] load local bake definitions
#1 reading from stdin 545B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 360B 0.0s done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.10-slim
#3 DONE 2.5s

#4 [internal] load .dockerignore
#4 transferring context:
#4 transferring context: 2B 0.0s done
#4 DONE 0.5s

#5 [1/5] FROM docker.io/library/python:3.10-slim@sha256:c1e4e6c01eb489c422288b2de34b0761ca316f7a2d98e2c33f47659a73ed108a
#5 resolve docker.io/library/python:3.10-slim@sha256:c1e4e6c01eb489c422288b2de34b0761ca316f7a2d98e2c33f47659a73ed108a
#5 resolve docker.io/library/python:3.10-slim@sha256:c1e4e6c01eb489c422288b2de34b0761ca316f7a2d98e2c33f47659a73ed108a 1.2s done
#5 DONE 1.5s

#6 [internal] load build context
#6 transferring context: 1.88kB 0.0s done
#6 DONE 1.4s

#5 [1/5] FROM docker.io/library/python:3.10-slim@sha256:c1e4e6c01eb489c422288b2de34b0761ca316f7a2d98e2c33f47659a73ed108a
#5 sha256:60e855e0dd9540cc6da6057297f1ab688e602e650a6075fc14da2c2f9c8884d1 0B / 1.29MB 0.2s
#5 sha256:acf9e06c2d45a94472a7d893455c898eff1f037b1c86b3e32f2dc67c57418da7 0B / 251B 0.2s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 0B / 13.88MB 0.2s
#5 sha256:acf9e06c2d45a94472a7d893455c898eff1f037b1c86b3e32f2dc67c57418da7 251B / 251B 0.7s done
#5 sha256:60e855e0dd9540cc6da6057297f1ab688e602e650a6075fc14da2c2f9c8884d1 1.29MB / 1.29MB 0.8s done
#5 extracting sha256:60e855e0dd9540cc6da6057297f1ab688e602e650a6075fc14da2c2f9c8884d1
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 1.05MB / 13.88MB 1.1s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 2.10MB / 13.88MB 1.2s
#5 extracting sha256:60e855e0dd9540cc6da6057297f1ab688e602e650a6075fc14da2c2f9c8884d1 0.3s done
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 3.15MB / 13.88MB 1.7s
```

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL
PS F:\SEM_7\Devops_Lab\flask-postgres-app> docker-compose up -d --build
#5 extracting sha256:60e855e0dd9540cc6da6057297f1ab688e602e650a6075fc14da2c2f9c8884d1 0.3s done
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 3.15MB / 13.88MB 1.7s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 4.19MB / 13.88MB 1.8s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 7.34MB / 13.88MB 2.0s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 8.39MB / 13.88MB 2.1s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 9.44MB / 13.88MB 2.3s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 11.53MB / 13.88MB 2.4s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 13.88MB / 13.88MB 2.7s
#5 sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 13.88MB / 13.88MB 2.9s done
#5 extracting sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c
#5 extracting sha256:2f7d10ea5bd937f48518553448d94b63ca5cca4a921cf96a374dee6d36b17c03c 1.8s done
#5 DONE 6.3s

#5 [1/5] FROM docker.io/library/python:3.10-slim@sha256:c1e4e6c01eb489c422288b2de34b0761ca316f7a2d98e2c33f47659a73ed108a
#5 extracting sha256:acf9e06c2d45a94472a7d893455c898eff1f037b1c86b3e32f2dc67c57418da7
#5 extracting sha256:acf9e06c2d45a94472a7d893455c898eff1f037b1c86b3e32f2dc67c57418da7 0.6s done
#5 DONE 6.9s

#7 [2/5] WORKDIR /app
#7 DONE 3.3s

#8 [3/5] COPY requirements.txt .
#8 DONE 0.2s

#9 [4/5] RUN pip install --no-cache-dir -r requirements.txt
#9 3.561 Collecting Flask==3.0.0
#9 3.659 Downloading flask-3.0.0-py3-none-any.whl (99 kB)
#9 3.701 _____ 99.7/99.7 kB 16.2 MB/s eta 0:00:00
#9 4.166 Collecting psycpg2-binary==2.9.9
#9 4.196 Downloading psycpg2_binary-2.9.9-cp310-cp310-manylinux_2_17_x86_64_manylinux2014_x86_64.whl (3.0 MB)
#9 4.379 _____ 3.0/3.0 MB 19.3 MB/s eta 0:00:00
#9 4.653 Collecting blinker>=1.6.2
#9 4.665 Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
#9 4.718 Collecting click>=8.1.3
#9 4.734 Downloading click-8.4.2-py3-none-any.whl (119 kB)
#9 4.742 _____ 119.2/119.2 kB 23.0 MB/s eta 0:00:00
#9 4.768 Collecting itsdangerous>=2.1.2
#9 4.786 Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
#9 4.872 Collecting Werkzeug>=3.0.0
#9 4.914 Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
```

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL powershell + - - - - -
PS F:\SEM_7\Devops_Lab\flask-postgres-app> docker-compose up -d --build
#9 4.872 Collecting Werkzeug>=3.0.0
#9 4.914 Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
#9 4.941 ----- 226.5/226.5 kB 9.1 MB/s eta 0:00:00
#9 4.982 Collecting Jinja2>=3.1.2
#9 5.222 Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
#9 5.233 ----- 134.9/134.9 kB 16.8 MB/s eta 0:00:00
#9 5.551 Collecting MarkupSafe>=2.0
#9 5.790 Downloading markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
#9 5.875 Installing collected packages: pycpg2-binary, MarkupSafe, itsdangerous, click, blinker, Werkzeug, Jinja2, Flask
#9 7.335 Successfully installed Flask-3.0.0 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.8 blinker-1.9.0 click-8.4.2 itsdangerous-2.2.0 pycpg2-binary-2.9.9
#9 7.337 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual
environment instead: https://pip.pypa.io/warnings/venv
#9 7.463
#9 7.463 [notice] A new release of pip is available: 23.0.1 -> 26.1.2
#9 7.463 [notice] To update, run: pip install --upgrade pip
#9 DONE 8.0s

#10 [5/5] COPY . .
#10 DONE 0.1s

#11 exporting to image
#11 exporting layers
#11 exporting layers 1.5s done
#11 exporting manifest sha256:b9f66f1d19674a439335d73a754c3a69ef7b1469aca61fae41237200ea38b0a 0.0s done
#11 exporting config sha256:0d643e0f0b13f7d7e2e31d9502ba5648f1f51f9accb8580351943c583f9b108b 0.0s done
#11 exporting attestation manifest sha256:cf259964d9c73e58ec22ff34123149dc878bc7a6527397ba352fbc7c5efa124 0.1s done
#11 exporting manifest list sha256:83785c9fe371fd43ce8f1ca03ed2ef8df5d69af0256abaaab5b4562ba1804f7c 0.0s done
#11 exporting manifest list sha256:83785c9fe371fd43ce8f1ca03ed2ef8df5d69af0256abaaab5b4562ba1804f7c 0.0s done
#11 naming to docker.io/library/flask-postgres-app-web:latest done
#11 unpacking to docker.io/library/flask-postgres-app-web:latest
#11 unpacking to docker.io/library/flask-postgres-app-web:latest 1.2s done
#11 DONE 3.0s

[+] up 21/21g provenance for metadata file
✓ Image postgres:latest Pulled 25.2s
✓ Image flask-postgres-app-web Built 26.6s
✓ Network flask-postgres-app_default Created 0.0s
✓ Volume flask-postgres-app_postgres-data Created 0.0s
✓ Container flask-postgres-app-db-1 Started 1.5s
✓ Container flask-postgres-app-web-1 Started 1.4s

Ln 28, Col 39 Spaces: 4 UTF-8 CRLF {} Python Python 3.10 (64-bit) Go Live Prettier
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL powershell + - - - - -
PS F:\SEM_7\Devops_Lab\flask-postgres-app> docker-compose up -d --build
✓ Container flask-postgres-app-web-1 Started 1.4s
PS F:\SEM_7\Devops_Lab\flask-postgres-app> docker-compose ps
time="2026-07-22T13:57:43+05:30" level=warning msg="F:\SEM_7\Devops_Lab\flask-postgres-app\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
flask-postgres-app-web-1 flask-postgres-app-web "python app.py" web 7 seconds ago Up 6 seconds 0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp
PS F:\SEM_7\Devops_Lab\flask-postgres-app>
```